

HYDRA Documentation

HYDRA MW4.0pe Windows Server 2016 Preparation Guide for DMC

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1 Introduction

This configuration manual explains how to configure Windows Server 2016 for use as a server for MPDVs HYDRA Dynamic Manufacturing Control (DMC) software in a HYDRA MW4.0pe environment.

The configuration of the server should always be based on MPDVs hardware and software recommendations for HYDRA MW4.0pe systems.

See manuals: **HW_SW_GUIDE.pdf**

!ATTENTION!

The Server must be a dedicated server for running HYDRA Dynamic Manufacturing Control (DMC). There must be no additional functionality used, e.g. like File Server, Print Server, OPC Server, Domain Controller (PDC, BDC), Active Directory Controller or similar.

If the configuration of your HYDRA DMC server deviates from the configuration described in this manual, additional efforts and expenses might be necessary for the DMC installation.

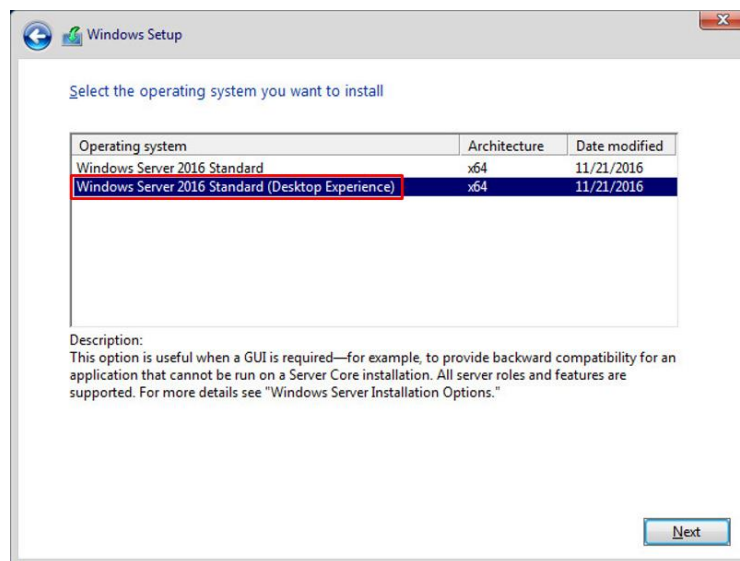
Already agreed shipping or installation appointments might be delayed or may have to be postponed.

HYDRA DMC might not be able to work properly on such a server.

It might even be impossible to perform a successful HYDRA DMC installation on such a server.

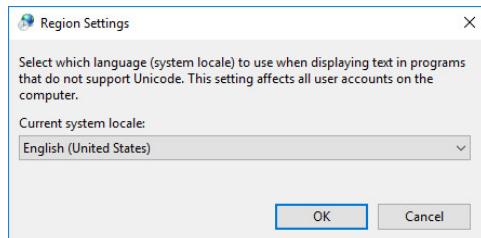
2 Operating System

- Server platform according to the hardware and software recommendations of MPDV Mikrolab GmbH is required.
- The Windows server must be a **dedicated server** for running HYDRA DMC.
There must be no additional functionality be used, e.g. like Domain Controller (PDC, BDC), Active Directory Controller, File Server, Print Server, OPC Server or similar.
In such cases it might be impossible to perform a successful installation of HYDRA DMC.
Pre installation tasks at a MPDV site might also not be possible or only with additional effort and expenses.
- Windows Server 2016 with a graphic user interface (**GUI**) is required.
During the installation of Windows Server 2016 select the option "**Desktop Experience**":



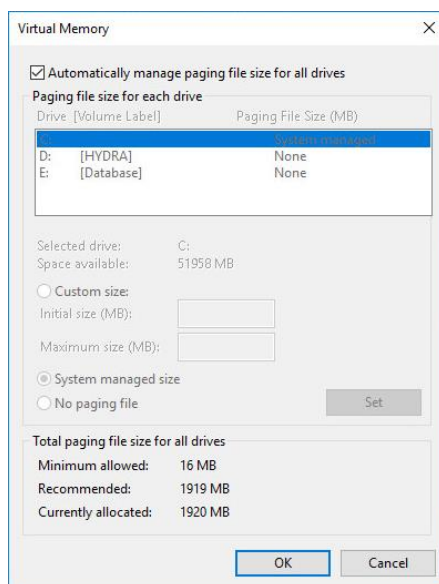
- The Windows desktop resolution should be set to **1920 x 1080** at least to **1280 x 1024**.
- Installed **.NET Framework 4.5.2** or higher.
- The recent Windows Service Pack approved by MPDV Mikrolab GmbH for HYDRA DMC must be installed on the server. See manual for hardware and software recommendations.
- All partitions must be formatted with file system **NTFS**.
See chapter "[3 Hard Disk Layout](#)" for more detailed information.

- Windows must be installed either with Desktop language **English** or **German**.
- The system locale of Windows must either be set to **“English (United States)”** or **“German (Germany)”**:



- **Virtual Memory**

Automatically manage paging file size for all drives (= Windows Default, recommended by MPDV)



If you choose to use individual settings, please make sure to follow the recommendations of the operating systems manufacturer to ensure a stable and performant system.

- **Network**

TCP/IP network protocol (**IPv4**) based on a Ethernet network must be available.

Default port used by HYDRA DMC: 3270 (if necessary that port can be changed).

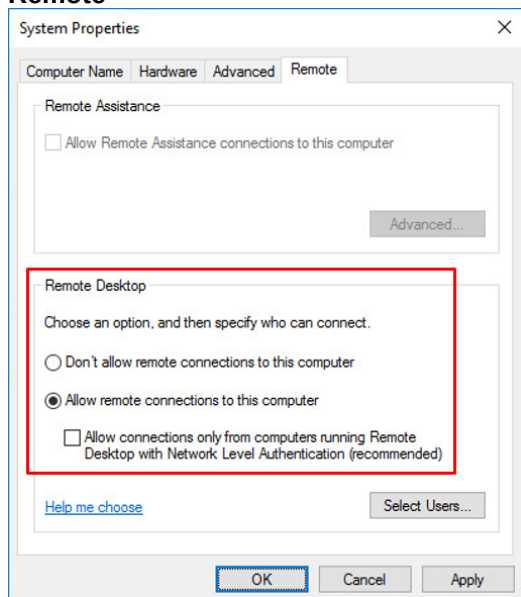
Depending on the extent of the DMC installation additional ports might be necessary.

- Choose any IP address and hostname for the server which suits your needs.
It is necessary that the HYDRA DMC server uses a **dedicated IP address**.

- Allow remote connections to the server:

Start-Button – Control Panel – System and Security – System – Advanced system settings –

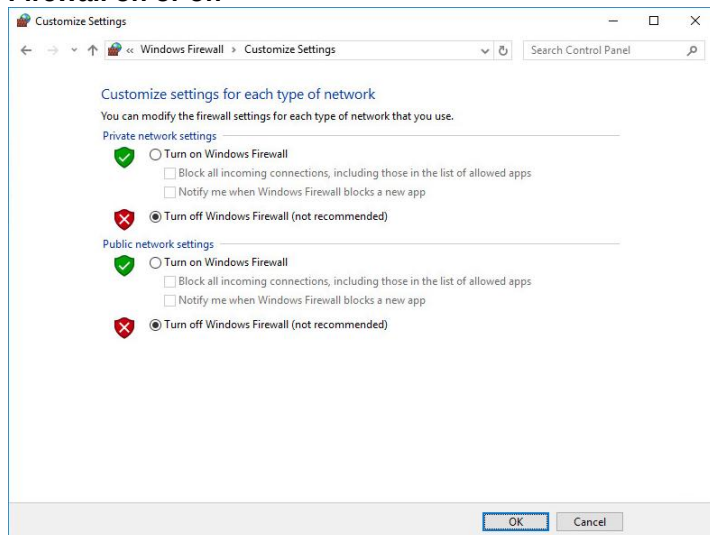
Remote



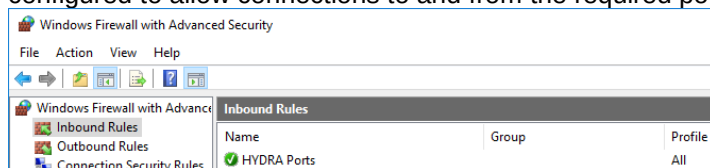
- The local Windows Firewall should be turned off:

Start-Button – Control Panel – System and Security – Windows Firewall – Turn Windows

Firewall on or off



If the local firewall needs to stay turned on there must be appropriate **Inbound** and **Outbound** rules configured to allow connections to and from the required ports of HYDRA (see above):

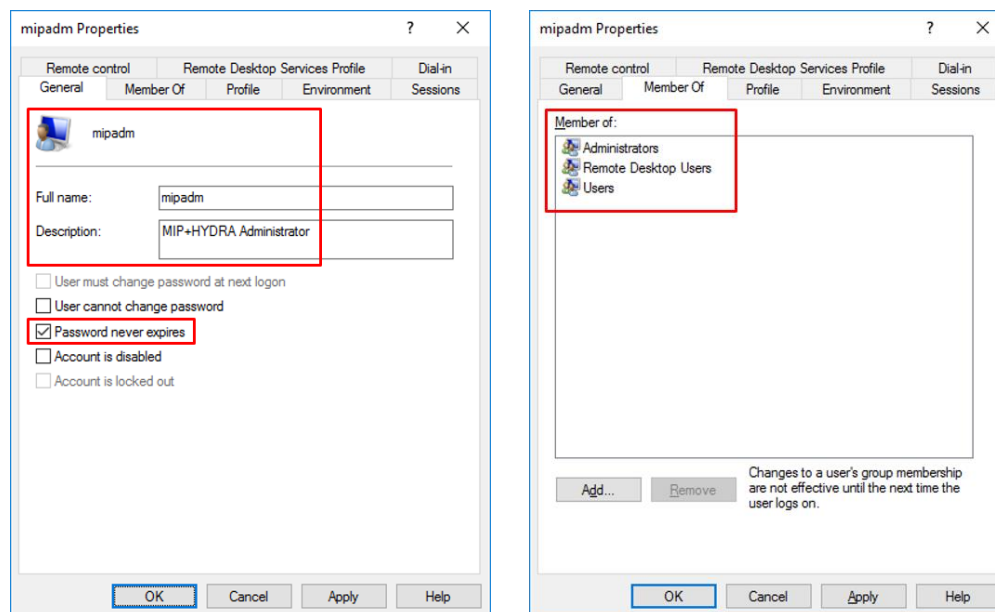


- **User mipadm**

There should be a local user account "**mipadm**" with full name "**mipadm**" and description "**MIP+HYDRA Administrator**" (see screenshots).

Alternatively to a local user account it would be possible to provide a domain user account "**mipadm**" in the customers Active Directory Domain Services.

User "mipadm" must be at least a member of the local groups "**Administrators**", "**Remote Desktop Users**" and "**Users**".



The password for user "mipadm" must never expire and should be set to "**Mip74821**".

Optionally any other password can be used which must then be disclosed to the MPDV employees who are installing HYDRA.

The following characters are **not** allowed for the password:

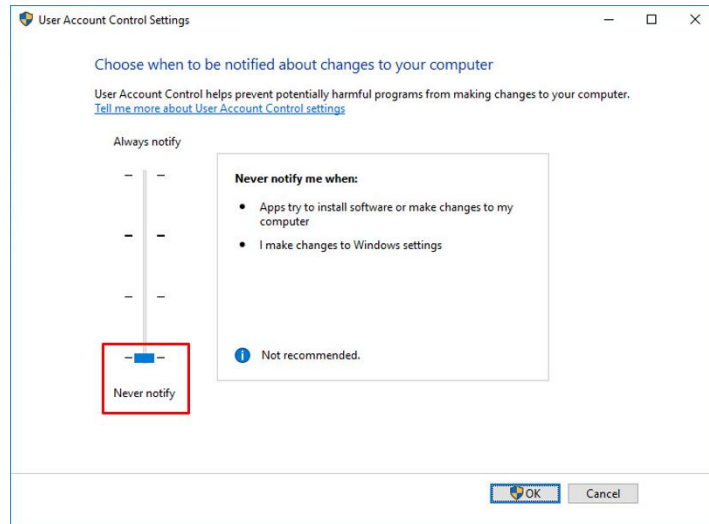
- Characters with ASCII Code > 126 (e.g. German umlauts, French "umlauts", etc.)

- Pipe: |

Periodic changes of the user's password would be possible as long as you take into account that there might be services running on the server using the login credentials for user "mipadm".

- User Account Control (UAC) **must** be deactivated.

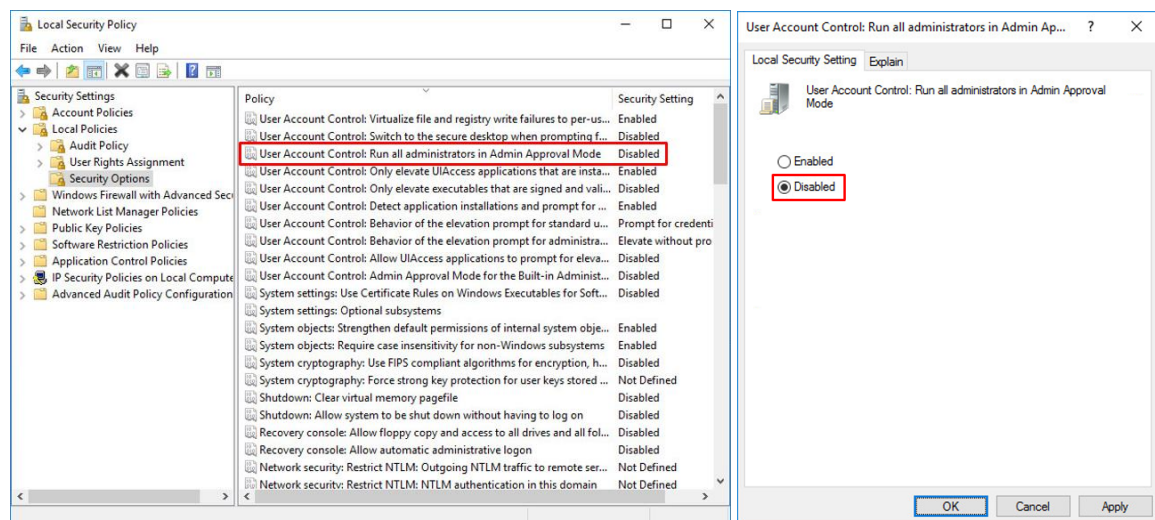
Start-Button – Control Panel – User Accounts – User Accounts – Change User Account Control settings



Never notify

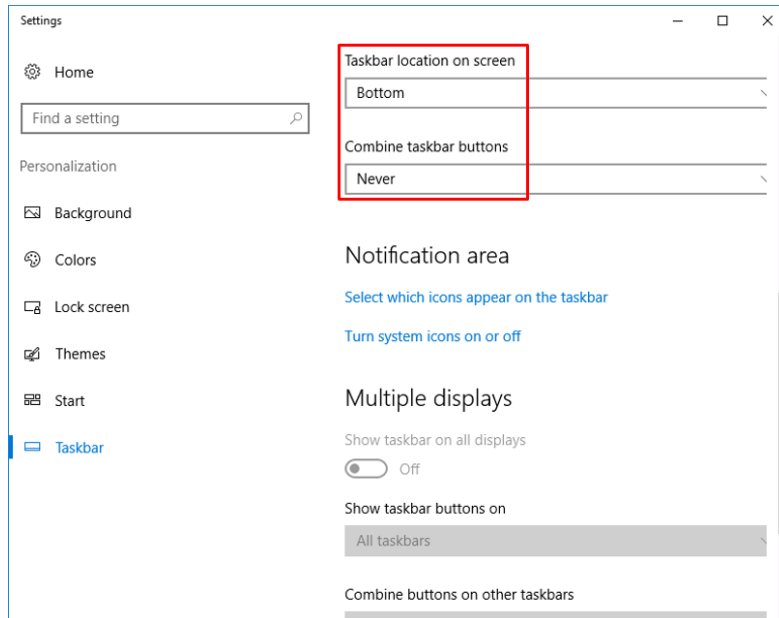
- In "Local Security Policy" the following policy **must** be disabled:

Start-Button – Windows Administrative Tools – Local Security Policy – Local Policies – Security Options – User Account Control: Run all administrators in Admin Approval Mode



Disabled

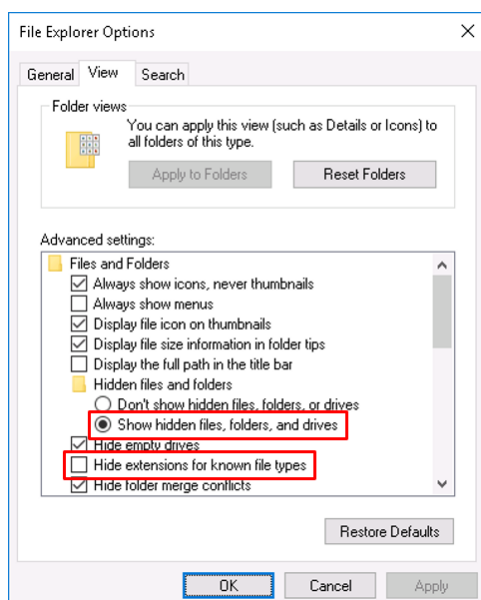
- Logon to the HYDRA DMC server as local user "mipadm".
- **Start-Button – Control Panel – Appearance and Personalization – Taskbar and Navigation**



Taskbar location on screen: Bottom

Combine taskbar buttons: Never

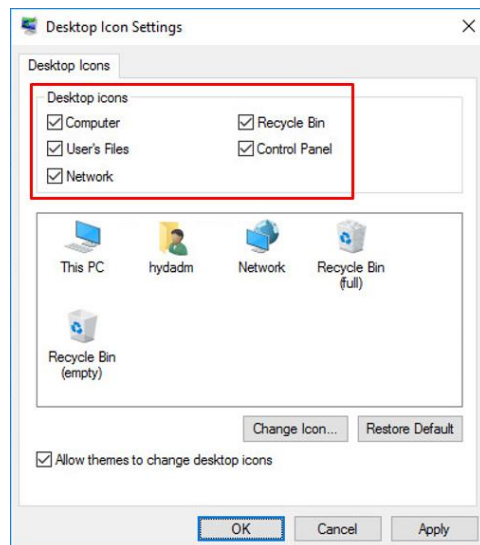
- **Start-Button – Control Panel – Appearance and Personalization – File Explorer Options – View**



Activate: Show hidden files, folders and drives

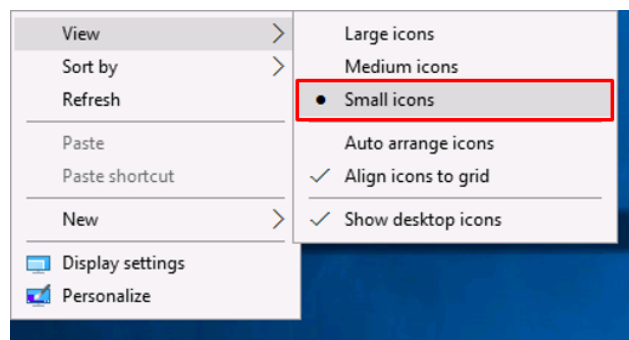
Deactivate: Hide extensions for known file types

- **Start-Button – Control Panel – Appearance and Personalization – Personalization – Change desktop icons**



Activate all Desktop icons

- Right click on the Windows desktop and then in "View" select: **"Small icons"**



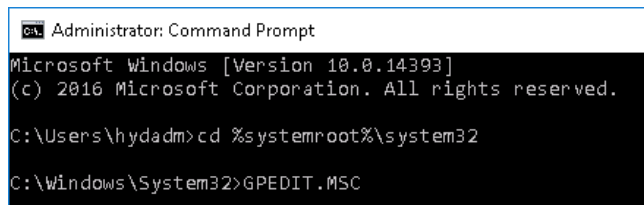
- Allow multiple Remote Desktop Sessions per User (optionally if required)

#Start-Button – Windows System – Command Prompt

type in the following commands:

```
cd %systemroot%\system32
```

```
GPEDIT.MSC
```



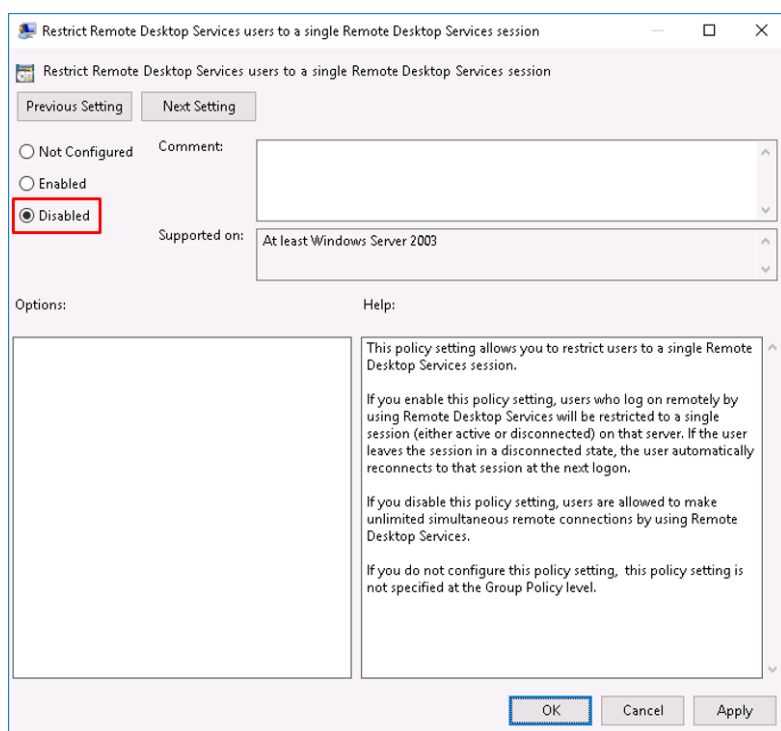
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\hydadm>cd %systemroot%\system32

C:\Windows\System32>GPEDIT.MSC
```

In the Local Group Policy Editor make the following settings:

Local Computer Policy – Computer Configuration – Administrative Templates – Windows Components – Remote Desktop Services – Remote Desktop Session Host – Connections – Restrict Remote Desktop Services users to a single Remote Desktop Services session



Disabled

3 Hard Disk Layout

The size of the available hard disk(s) installed depends on the recent recommendations made by MPDV Mikrolab GmbH for certain HYDRA Dynamic Manufacturing Control (DMC) system sizes.

There must be at least the following two partitions available: **C:** and **D:**.

Depending on Windows and the available amount of RAM drive **C:** should be **at least 80-100 GB** in size.

Depending on the dedicated DMC system size drive **D:** must provide **100 – 500 GB** of available disk space.

Available CD/DVD drives should use a drive letter at the end of the available range, e.g.: **Z:**.

All partitions must be formatted with file system **NTFS**.

Configuration example:

C:\ NTFS	100 GB	Windows, tools and pagefile.sys
D:\ NTFS	500 GB	DMC application and DMC data and archive files